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What is Artificial Intelligence?
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** Artificial Intelligence is a branch of Data Science and Computer Science.
** AI aims to create machines or systems that can think and act like humans.
** AI is used to create intelligent machines and intelligent software systems.

** AI means making computers perform tasks that normally require human intelligence.**
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##### AI Real-Life Examples #####
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** Artificial Intelligence is used in many real-time applications, such as:

    1. Self-driving cars
    2. Fraud detection systems in banks
    3. Spam detection in emails
    4. ChatGPT answering questions and generating images
    5. GitHub Copilot generating code
    6. Voice assistants like Siri, Alexa, and Google Assistant
    7. Product recommendations in Amazon and Flipkart
    8. Movie recommendations in Netflix and YouTube
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What is Machine Learning?
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** Machine Learning is a subset of Artificial Intelligence.

** Machine Learning allows computers to learn patterns from data without writing programs
explicitly for every condition.

## Simple Definition

**Machine Learning is a technique where the computer learns from data and makes predictions or
decisions.**
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### Traditional Programming:
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    Input + Rules → Output
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### Machine Learning:
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Input + Output Data → Learning Algorithm → Model New Input → Trained Model → Prediction
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## Example
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==> Suppose we want to calculate whether a student will pass.
```

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if attendance > 75 and marks > 40:
    PASS
else: FAIL

```

Here, we manually create the rules.

**** In Machine Learning, we provide historical student data:**

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--- Hours studied
--- Attendance
--- Preious Marks
--- Assignments reports

```

**** The algorithm learns patterns from the historical data.**

New Student Data ---> ML Model --> PASS/FAIL

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Why Machine Learning?
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**** Traditional software requires developers to define rules.**

**** For some problems, rules are very difficult to define.**

Example:

Spam Email Detection

It is difficult to write rules such as:

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if email contains "offer" or "buy back":
    spam

```

**** instead of writing these conditions manually, we can provide historical emails for ML algorithm then it will learn patterns from the data and it can identify SPAM emails easily.**

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Types of Machine Learning
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1) Supervised learning

2) Unsupervised learning

3) Reinforcement learning

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Supervised Learning
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=> In supervised learning, the training data contains both:

input + Correct Output

Example:

Student Information â†’ Pass/Fail

House Features → House Price

Email → Spam/Not Spam

** The inputs are called FEATURES.

** The output is called the TARGET.

=> Supervised learning is mainly divided into:

1) Regression

2) Classification

** Regression is used when the target is a continuous numerical value.

Ex: house price, salary, temperature, sales, revenue, stock price prediction

** Classification is used when the output belongs to a category.

Ex: spam or not spam, pass or fail, Fraud or Not Fraud, Disease or No Disease

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Unsupervised Learning
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** In unsupervised learning, we do not have a target/output column.

** The algorithm tries to discover patterns or groups in the data.

Example:

Customer data:

Age
Income
Spending Score

The algorithm may discover:

Group 1 → Low spending customers

Group 2 → Medium spending customers

Group 3 → High spending customers

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Reinforcement Learning
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** In Reinforcement Learning, an agent/model learns by interacting with an environment.

** The agent performs an action and receives:

Reward or Penalty

**** A Reward is feedback given to the agent after it performs an action.**

**** A Penalty is negative feedback to the agent / model.**

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Machine Learning Workflow
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- 1) Understand business problem
- 2) Collect Data
- 3) Understand Data
- 4) Clean Data
- 5) EDA (Exploratory Data Analysis)
- 6) Feature Engineering (Data Transformation)
- 7) Train and Test Split (Separate training data and testing data)
- 8) Pre Processing (convert data into numerical format)
- 9) Select Algorithm
- 10) Train Model
- 11) Make Predictions
- 12) Evaluate Model
- 13) Tune Model
- 14) Save model (ex : house_price_model.pkl)
- 15) Deploy Model

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Important ML terminology
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1) Data Set

=> Collection of data used for analysis and used for training & testing

Ex:

=> Below is the students data set

Study hours
Attendance
Previous marks
Result (Pass or Fail)

Syntax :

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import pandas as pd
```

```
df = pd.read_csv("students.csv")
```

=> Here dataset is the complete table

Features = Study Hours, Attendance, Previous marks

Target = Result

=> Features are input variables used by the Machine Learning model.

=> Target is the value that we want the model to predict.

Features = Area, Bedrooms, Age

Target = Price

=> In Dataset, X represents input values and Y represents Output values

2) Training Data and Testing data

=> We should not train and test the model using the same data.

=> Why because the model could simply memorize the data.

=> We divide data into 80 - 20 formula

Training data 80%

Testing data 20%

Example:

Dataset = 1000 records

800 records â†’ Training

200 records â†’ Testing

=> The model learns patterns from the training data.

3) random_state

=> When we split the dataset, records are randomly selected.

=> If we execute the program multiple times, the split can change.

random_state = 10

Note : The important point is that we use a fixed value for reproducibility.

4) Data Preprocessing

=> Real-world data is usually not clean.

Examples :

1) Missing Values

2) Duplicate Records

3) Incorrect Values

Therefore, we need preprocessing.

5) Mean vs Median

Mean = $\text{sum_of_values} / \text{total_num_of_values}$

Median = Middle value after sorting the data

6) Categorical Data

=> Machine Learning algorithms generally require numerical inputs.

Example:

Gender

Male

Female

=> We can convert categorical data into numerical representation.

7) Encoding

Label Encoding

Low ----> 0

Medium ---> 1

High ----> 2

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ML Algorithms
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** Regression : It is used when we want to predict continuous numeric value

Ex :

- 1) House Price Prediction
- 2) Salary Prediction
- 3) Sales Prediction
- 4) Temperature Prediction
- 5) Stock Price Prediction
- 6) Revenue Prediction

** Common Regression Algorithms

- 1) Linear Regression
- 2) Multiple Linear Regression
- 3) Polynomial Regression

- 4) Decision Tree Regression
- 5) Random Forest Regression
- 6) Support Vector Regression (SVR)

**** Classification :** It is used when we want to predict a category

Ex:

- 1) Spam Detection
- 2) Disease Detection
- 3) Fraud Detection
- 4) Loan Approval
- 5) Sentiment Analysis
- 6) Image Classification
- 7) Student Pass/Fail prediction

**** Common Classification Algorithms**

- 1) Logistic
- 2) Decision Tree Classifier
- 3) Random Forest Classifier
- 4) Support Vector Machine (SVM)
- 5) KNN (Nearest Neighbors)
- 6) Naive Bayes

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Assignment-1 : Develop ML model based on Linear Regression

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